Drawbar Organ Emulation with DX7 Patches „888“ and „888 888“

Since I was dissatisfied with the organ sounds I had previously created for miniDexed, I took a closer look at the Hammond organ today and built three performances (004_Mirage:058-060), implemented as follows. The two DX7 voice patches **„888“** (used on TG1) and **„888 888“** (used on TG4) together recreate the nine drawbars of a classic tonewheel organ, plus key-click and a warm, tube-like saturation. By combining them in a DreamDexed Performance and using **TG-Link**, you can blend the lower and upper drawbar ranges directly from Dexed’s Performance page.

The two patches are printed in the **Appendix** at the end of this document.
Note that in the **„888 888“** patch, all six **Output Levels** are set to **86 –** this gives a balanced, full-bodied tone as a starting point; lowering individual operators darkens the corresponding drawbar just like pushing it in.

Operator → Drawbar Mapping

Lower Drawbars (16′, 5 1/3′, 8′) – Voice „888“

Drawbar	Pitch Ratio	Operator	Note
16′	0.5	OP1	Pure sine carrier (no modulation)
5 1/3′	1.5	OP2	Fine-tuned to +50 cent ≈ 1.50
8′	1.0	OP3	Unison

•**Operator 4** provides the **key-click** (sharp attack, ratio 3.0).
•**Operators 5 & 6** create the **tube/smoky warmth** (ratios 1.1 and 4.2, combined through FM modulation).

In Algorithm 31, each of OP1, OP2 and OP3 is an independent carrier whose **Output Level** directly sets the volume of the corresponding drawbar.
The modulating operators (OP6 → OP5) addsaturation and Operator 4 add the click, but the fundamental drawbar balance is determined purely by OP1-3.

Upper Drawbars (4′ – 1′) – Voice „888 888“

Drawbar	Pitch Ratio	Operator
4′	2.0	OP1
2 2/3′	3.0	OP2
2′	4.0	OP3
1 3/5′	5.0	OP4
1 1/3′	6.0	OP5
1′	8.0	OP6

All operators are routed in Algorithm 32 (parallel carriers), meaning each operator is a pure sine wave at its given pitch – exactly like a tonewheel.
Here too, the **Output Level** of each operator is the drawbar volume for that pitch.

Performance Setup (DreamDexed)

The example performance **000060_888 888 888** groups the voices as follows:

TG	Voice	MIDI Ch.	TG-Link	Volume	Purpose
1	„888“	1	A	22	Lower drawbars + click + tube
2	„888“	1	A	22	(doubles TG1 for stereo)
3	„888“	1	A	22	(doubles TG1)
4	„888 888“	1	B	18	Upper drawbars
5	„888 888“	1	B	18	(doubles TG4 for stereo)
6-8	muted	0	–	–	Not used

•**TG-Link A** (TG1-3) controls the lower drawbar section.
•**TG-Link B** (TG4-5) controls the upper drawbar section.
•All linked TGs respond to the same MIDI channel, so you play them like a single instrument.

How to “Pull” the Drawbars

1. Individual Drawbar Control via Operator Output Levels (essential)

The real power of this drawbar emulation lies in editing the voices themselves:

•Open **Voice „888“** in Dexed.

•**OP1 Output Level** adjusts the **16′** drawbar.
•**OP2 Output Level** adjusts the **5 1/3′** drawbar.
•**OP3 Output Level** adjusts the **8′** drawbar.
•OP4-6 handle click/warmth – lower their levels if you want a clean tone.
•Open **Voice „888 888“** in Dexed.

•**OP1 Output Level** adjusts the **4′** drawbar.
•**OP2 Output Level** adjusts the **2 2/3′** drawbar.
•… and so on up to OP6 for the **1′** drawbar.

Changes to these operator levels immediately affect the tone, just like pulling a physical drawbar. Because the voices are stored inside the performance, any adjustment is permanent for that patch – you can save the modified voices as new cartridge entries and re-assign them to the TGs.

2. Quick Section Balance with TG-Link Volumes

On the Performance page you have two master §drawbar-group-faders§:

•**TG Volume 1** (linked A) controls the overall level of the lower three drawbars plus the click/tube artefacts.
•**TG Volume 4** (linked B) controls the overall level of the upper six drawbars.

Use these to quickly match the lower and upper halves, while the fine drawbar-by-drawbar character is set via the operator levels.

3. Stereo Width

The doubled TGs (TG2, TG3, TG5) with different pan settings create a wider stereo image. You can delete the duplicates if mono is sufficient.

4. Removing the Click/Warmth

Edit voice **„888“** in Dexed:

•Set **OP4** output level to 0 (removes the click).
•Lower or mute **OP5** and **OP6** to eliminate the tube/smoky effect.

You can then save a „clean“ version of the lower drawbar voice.

5. Extreme Flexibility (Advanced)

If you prefer to control every drawbar from the Performance page without editing voices, you can split each drawbar into its own TG:

•Create a voice with Algorithm 32 where *only one operator* is active (e.g., OP1 alone).
•Place each such voice on a separate TG.
•Set the volume of each TG to its desired drawbar level.

This gives you real-time faders for each of the nine pitches, at the cost of using more TGs.

Summary

This elegant DX7-based drawbar engine uses two specially crafted voices that map all nine traditional tonewheel pitches onto six operators each, leaving room for expressive artefacts. The primary “drawbar” knobs are the **Output Levels of the carriers** in Dexed’s voice editor. Combined with TG-Link and Performance volume controls, you get fast, dynamic access to the classic organ character – directly from Dexed or DreamDexed’s interface.

Appendix:

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DX7 VOICE DATA SHEET by SOUNDPLANTAGE.COM

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Performance: 004:060 888 888 888 - TG1 - 888

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ALGORITHM: 31 FEEDBACK: 6 TRANSPOSE: 0

PARAM	OP1	OP2	OP3	OP4	OP5	OP6
OSC MODE	RATIO	RATIO	RATIO	RATIO	RATIO	RATIO
COARSE	0.5	1.0	1.0	3.0	1.0	4.0
FINE (FREQ)	0.50	1.50	1.00	3.00	1.10	4.20
DETUNE	+0	+0	+0	+0	+0	+0
AMP MOD SENS	0	0	0	0	3	0
VEL SENS	0	0	0	0	0	0
OUT LEVEL	99	89	99	89	51	74
BREAK POINT	A-1	A-1	A-1	A-1	A-1	A-1
L DEPTH	0	0	0	0	0	0
R DEPTH	0	2	0	1	4	9
L CURVE	-LIN	-LIN	-LIN	-LIN	-LIN	-LIN
R CURVE	-EXP	-EXP	-EXP	-EXP	-EXP	-EXP
RATE SCALING	0	0	0	0	0	0
EG LEVEL 1	99	99	99	99	99	99
EG LEVEL 2	99	99	99	99	99	99
EG LEVEL 3	99	99	99	0	99	99
EG LEVEL 4	0	0	0	0	0	0
EG RATE 1	77	71	81	86	73	99
EG RATE 2	99	99	99	99	99	99
EG RATE 3	99	99	99	65	99	99
EG RATE 4	99	99	99	99	99	33

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LFO, PITCH EG & SYNC

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LFO WAVE: TRIANGLE SPEED: 33 DELAY: 0

PMD: 19 AMD: 45 P MOD SENS: 1

LFO SYNC: OFF OSC SYNC: OFF

PITCH EG LEVEL: L1=50 L2=50 L3=50 L4=50

PITCH EG RATE : R1=99 R2=99 R3=99 R4=99

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DX7 VOICE DATA SHEET by SOUNDPLANTAGE.COM

Performance: 004:060 888 888 888 - TG4 - 888 888

ALGORITHM: 32 FEEDBACK: 0 TRANSPOSE: 0

PARAM	OP1	OP2	OP3	OP4	OP5	OP6
OSC MODE	RATIO	RATIO	RATIO	RATIO	RATIO	RATIO
COARSE	2.0	3.0	4.0	5.0	6.0	8.0
FINE (FREQ)	2.00	3.00	4.00	5.00	6.00	8.00
DETUNE	+0	+0	+0	+0	+0	+0
AMP MOD SENS	0	0	0	0	0	0
VEL SENS	0	0	0	0	0	0
OUT LEVEL	86	86	86	86	86	86
BREAK POINT	A-1	A-1	A-1	A-1	A-1	A-1
L DEPTH	0	0	0	0	0	0
R DEPTH	3	3	6	10	12	14
L CURVE	-LIN	-LIN	-LIN	-LIN	-LIN	-LIN
R CURVE	-EXP	-EXP	-EXP	-EXP	-EXP	-EXP
RATE SCALING	0	0	0	0	0	0
EG LEVEL 1	99	99	99	99	99	99
EG LEVEL 2	99	99	99	99	99	99
EG LEVEL 3	99	99	99	99	99	99
EG LEVEL 4	0	0	0	0	0	0
EG RATE 1	99	99	99	99	99	99
EG RATE 2	99	99	99	99	99	99
EG RATE 3	99	99	99	99	99	99
EG RATE 4	99	99	99	99	99	99

LFO, PITCH EG & SYNC

LFO WAVE: TRIANGLE SPEED: 22 DELAY: 0
PMD: 0 AMD: 67 P MOD SENS: 1
LFO SYNC: ON OSC SYNC: OFF

PITCH EG LEVEL: L1=50 L2=50 L3=50 L4=50
PITCH EG RATE : R1=99 R2=99 R3=99 R4=99